

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Determining whether the code generated by AI works correctly through testing and debugging for further evaluation.
2. An understanding of stakeholder goals and Python syntax, ensuring the software adheres to the proposed requirements and functions as intended.
3. Ensuring user information is kept safe through security implementations, making sure that the code generated leaves no vulnerabilities for users and adjusting/implementing code appropriately to ensure safety.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	Not necessarily	It is programming as code was developed, but it is not necessarily software engineering as the application does not address a wider audience, therefore not much consideration needs to be made if the application is intended to solve a small

			problem, presumably for personal use.
B	Yes	Yes	The intention of the application is to be used by many users (employees), in which this scenario therefore involves programming and software engineering as the system requires designing, testing etc. to ultimately provide services to a large audience.
C	Yes	No	Although generated by AI, it is still considered as programming as it is creating code based on requirements described in the prompt. However, it is not necessarily software engineering as the scenario implies a lack of consideration of stakeholder considerations, system layouts and etc. hence, only a <i>simple</i> application is created

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Patients	Request appointment bookings, have accurate patient and appointment information
Healthcare Practitioners	View patient information and appointment details
Receptionists	Create, edit and view patient information and appointments
System administrator	Maintain software functionality, user data and system access
Manager	Moderate clinic operations and ensure patients' recorded information are correct

Task 2 - Identify current problems

1. The clinic uses a combination of digital and physical processes to document patient and appointment information, which ultimately causes issues in locating information, duplicate bookings and inconsistent appointment status information
2. Manual processes (e.g. bookings, operational reports, cancellations), can create mistakes involving scheduling conflicts.
3. Information can be entered more than once, causing duplicate patient records and inconsistencies in information.
4. Disorganisation of patient records, slows down efficiency and can slow down processes if in need of specific patient/appointment information

Task 3 - Ask client questions

1. Who primarily manages practitioner availability? The practitioners themselves or the receptionists?
2. Who is allowed to make/edit appointment bookings or cancellations?
3. Can a practitioner have more than one appointment booked at the same time?
4. Do appointments remain in the scheduling history even if it has been cancelled?
5. After the implementation of the program, will the clinic still continue to use manual processes? Or will they fully integrate the program into their practices.

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes	Yes	Add as it is a core feature especially requested by the client in regards to their problem
Facial recognition login	No	No	Client can request the feature, but it is not what they explicitly wanted to add
AI diagnosis recommendations	No	No	Initially the feature can not guarantee an accurate diagnosis, and causes risks in reliability of the AI as well as the clinic, therefore do not add

Patient search	Yes	Possibly	Assuming this feature is for the receptionists, it can be a feature implemented, though it is not explicitly requested by the client therefore do not add unless requested
Online payment	No	No	Client does not state any issues regarding payments, therefore it is not necessary to add this feature
Practitioner schedule view	Possibly	Possibly	Requires practitioner's insights on the idea, therefore do not implement until confirmed with the client
Insurance processing	No	No	Client has no requirements regarding insurance processing, therefore do not add
Treatment-plan generation	No	No	Not within client's scope, do not add

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

Communication with stakeholders → Software engineers validate and analyse engineering and stakeholder requirements. In addition, stakeholders may not be programming professionals, therefore software engineers can suggest a scope of opportunities that can improve on the stakeholders ideas, mitigating risks and further clarify the blueprint of the project.